Presidential Graduate School
BSCIT 375: Knowledge Discovery and Data Science
Class Test
Time: 2 Hours **Full Marks: 50**

IMPORTANT EXAMINATION NOTICE

Please ensure you bring your own **statistics tables**, **log tables**, and **calculator** to the examination.

Note: Borrowing materials from others is ***strictly prohibited****. Students will not be allowed to share or request items from their peers during the session.*

Use of mobile phones is strictly prohibited. No exceptions, no excuses. **Phones out, exam over.**

- **Attempt any 5 (FIVE)** questions.
- Show all steps in your calculations.

Question 1: Univariate Inference **[10 Marks]**

(a) What is the difference between a point estimate and an interval estimate?
A sample of 5 API response times (in milliseconds) is: 12, 14, 13, 15, 16. Find the sample mean and sample standard deviation.
(Use: $\bar{x} = \frac{\sum x_i}{n}$, $s = \sqrt{\frac{\sum(x_i-\bar{x})^2}{n-1}}$) (2+3=5)

(b) A company claims that only 5% of its products are defective. In a random sample of 400 products, 24 are found defective. Test the claim at $\alpha = 0.05$ using a one-tailed test ($H_a : p > 0.05$). Compute the test statistic $Z = \frac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)/n}}$ and state your conclusion.
(Critical value for one-tailed test at $\alpha = 0.05$ is $Z_{0.05} = 1.645$) (5)

Question 2: Multivariate Analysis **[10 Marks]**

(a) State two assumptions of the two-sample t-test.
Two independent groups have the following summaries:
Group A: $n_1 = 30$, $\bar{x}_1 = 25$, $s_1 = 4$
Group B: $n_2 = 25$, $\bar{x}_2 = 28$, $s_2 = 5$
Calculate the t-statistic using Welch's formula: $t = \frac{\bar{x}_1-\bar{x}_2}{\sqrt{s_1^2/n_1+s_2^2/n_2}}$. (2+3=5)

(b) A researcher wants to know if customer satisfaction (Satisfied, Not Satisfied) is independent of the region (North, South). The observed frequencies are:

	Satisfied	Not Satisfied	Total
North	50	30	80
South	60	40	100
Total	110	70	180

Calculate the expected frequency for the cell "North and Satisfied" and compute the χ^2 statistic for that single cell using $\frac{(O-E)^2}{E}$. (You do not need to compute the full χ^2 value.)
(5)

Question 3: Data Preprocessing **[10 Marks]**

(a) What is the purpose of data normalization? Give one example of a normalization technique.
For the values [10, 20, 30, 40, 100], apply Min-Max normalization to scale them to the range [0,1]. Show the normalized value for 30. (2+3=5)

(b) Calculate the Pearson correlation coefficient r for the following two variables X and Y:

	X	Y
1	2	4
2	4	8
3	6	12

Use the formula $r = \frac{\sum(X_i-\bar{X})(Y_i-\bar{Y})}{\sqrt{\sum(X_i-\bar{X})^2 \sum(Y_i-\bar{Y})^2}}$. (5)

Question 4: Classification – Bayes and Decision Trees **[10 Marks]**

(a) What is the main assumption of the Naive Bayes classifier?
For the feature "Outlook" given class "Play", the frequencies are:

Outlook	Play = Yes	Play = No
Sunny	2	3
Overcast	4	0
Rainy	3	2

Estimate $P(\text{Sunny} \mid \text{Yes})$ using Laplace smoothing with $\alpha = 1$. (There are 3 possible Outlook values.) (2+3=5)

(b) A dataset has 8 instances: 5 "Yes" and 3 "No". A split on a feature produces two subsets: - Subset 1: 4 instances (3 Yes, 1 No) - Subset 2: 4 instances (2 Yes, 2 No) Calculate the Gini impurity of each subset and then the weighted Gini impurity of the split.
(Formula: $\text{Gini}(S) = 1 - \sum p_i^2$) (5)

Question 5: KNN and K-Means **[10 Marks]**

(a) Distinguish between supervised and unsupervised learning with one example of each. Given points $P_1(1,2)$, $P_2(3,4)$, $P_3(5,6)$. Using K-Means with $K = 2$ and initial centroids at P_1 and P_2, assign each point to the nearest centroid after the first iteration. Use Euclidean distance. (2+3=5)

(b) A KNN regression problem uses the following normalized data. Predict the price for a new house with normalized size = 0.6 and normalized bedrooms = 0.5 using $K = 3$ and Euclidean distance.

House	Size (norm)	Bedrooms (norm)	Price ($k)
1	0.0	0.0	200
2	0.4	0.5	240
3	0.6	0.0	260
4	0.8	0.5	300

New point: (0.6, 0.5) (5)

Question 6: Association Rules and Model Evaluation **[10 Marks]**

(a) Define support and confidence for an association rule $X \Rightarrow Y$.
In a transaction database, support($\{Milk\}$) = 0.5, support($\{Bread\}$) = 0.4, support($\{Milk, Bread\}$) = 0.3. Calculate the confidence and lift for the rule $\{Milk\} \Rightarrow \{Bread\}$.
(Lift = $\frac{\text{confidence}(X \Rightarrow Y)}{\text{support}(Y)}$) (2+3=5)

(b) A confusion matrix for a spam detection model is:

	Predicted: Spam	Predicted: Not Spam
Actual: Spam	80	20
Actual: Not Spam	10	90

Compute precision, recall, and F1-score.
(Precision = $\frac{TP}{TP+FP}$, Recall = $\frac{TP}{TP+FN}$, F1 = $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$) (5)

Good Luck!